

Terna Engineering College

Computer Engineering Department

Program: Sem VI

Course: Cloud Computing Lab (CSL605)

PART A

(PART A: TO BE REFERRED BY STUDENTS)

Experiment No.9

Title: To study and Implement Containerization using Docker

A.1 Objective: To know the basic differences between Virtual machine and Container. It involves demonstration of creating, finding, and building, installing, and running Linux/Windows application containers inside local machine or cloud platform.

A.2 Prerequisite:

Knowledge of Networking, Distributed Computing and knowledge of Software architectures.

A.3 Objective:

Objectives this experiment is to provide students an overview AWS, its Features and Services.

A.4 Outcome: (LO3)

After successful completion of this experiment student will be able to run Linux/ windows application containers inside local machine.

A.5 Theory :

Docker is the containerization platform that is used to package your application and all its dependencies together in the form of containers to make sure that your application works seamlessly in any environment which can be developed or tested or in production.

**Containerization:**

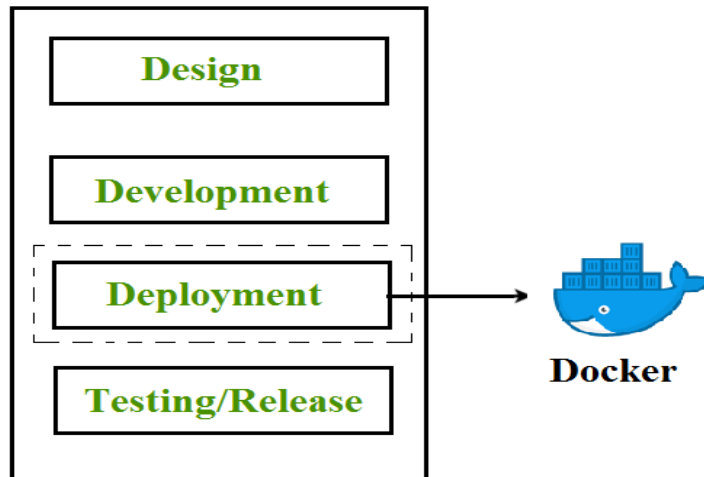
Containerization is OS-based virtualization that creates multiple virtual units in the userspace, known as Containers. Containers share the same host kernel but are isolated from each other through private namespaces and resource control mechanisms at the OS level. Container-based Virtualization provides a different level of abstraction in terms of virtualization and isolation when compared with hypervisors. Hypervisors use a lot of hardware which results in overhead in terms of virtualizing hardware and virtual device drivers. A full operating system (e.g -Linux, Windows) runs on top of this virtualized hardware in each virtual machine instance. But in contrast, containers implement isolation of processes at the operating system level, thus avoiding such overhead. These containers run on top of the same shared operating system kernel of the underlying host machine and one or more processes can be run within each container. In containers you don't have to pre-allocate any RAM, it is allocated dynamically during the creation of containers while in VMs you need to first pre-allocate the memory and then create the virtual machine. Containerization has better resource utilization compared to VMs and a short boot-up process. It is the next evolution in virtualization.

Containers can run virtually anywhere, greatly easy development and deployment: on Linux, Windows, and Mac operating systems; on virtual machines or bare metal, on a developer's machine or in data centers on-premises; and of course, in the public cloud. Containers virtualize CPU, memory, storage, and network resources at the OS level, providing developers with a sandboxed view of the OS logically isolated from other applications. Docker is the most popular open-source container format available and is supported on Google Cloud Platform and by Google Kubernetes Engine.

After you understand the concept of Docker, now let's get into the implementation. There are several steps that we will do:

1. Install Docker
2. Create a file called Dockerfile

3. Build the image
4. Run the image



Useful link:

<https://docs.docker.com/>

PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case there is no ERP access available)

Roll No.C54	Name: Quazi Md Moinuddin
Class :TE-C	Batch : C-3
Date of Experiment: 3-4-25	Date of Submission: 5-4-25
Grade :	

B.1 Question of Curiosity:

(To be answered by student based on the practical performed and learning/observations)

Q.1 What is containerization?

Ans: Containerization is a software deployment process that bundles an application's code with all the files and libraries it needs to run on any infrastructure. Traditionally, to run any application on your computer, you had to install the version that matched your machine's operating system. For example, you needed to install the Windows version of a software package on a Windows machine. However, with containerization, you can create a single software package, or container, that runs on all types of devices and operating systems.

Q.2 Difference between virtualization and containerization.

Sno.	Virtual Machines (VM)	Containers
1	VM is piece of software that allows you to install other software inside of it so you basically control it virtually as opposed to installing the software directly on the computer.	While a container is a software that allows different functionalities of an application independently.

2.	Applications running on VM system can run different OS.	While applications running in a container environment share a single OS.
3.	VM virtualizes the computer system.	While containers virtualize the operating system only.
4.	VM size is very large.	While the size of container is very light; i.e. a few megabytes.
5.	VM takes minutes to run, due to large size.	While containers take a few seconds to run.
6.	VM uses a lot of system memory.	While containers require very less memory.
7.	VM is more secure.	While containers are less secure.
8.	VM's are useful when we require all of OS resources to run various applications.	While containers are useful when we are required to maximise the running applications using minimal servers.
9.	Examples of VM are: KVM, Xen, VMware.	While examples of containers are: RancherOS, PhotonOS, Containers by Docker.

Q.3 What is docker image?

Ans : A Docker image is a lightweight, standalone, executable package that includes everything needed to run a piece of software, including the code, runtime, libraries, dependencies, and configurations. It is essentially a snapshot of a Docker container. Docker images are built from a Dockerfile, which contains instructions on how to assemble the image. Images are stored in a Docker registry, such as Docker Hub, where they can be shared and pulled by others. Docker images follow a layered approach, where each layer represents a specific instruction in the Dockerfile. These layers are stacked on top of each other, and when a new image is created, only the layers that have changed are rebuilt, making the process efficient and lightweight.

Q.4 What is docker container?

Ans : A Docker container is a lightweight, runnable instance of a Docker image. It encapsulates an application and its dependencies in an isolated environment, ensuring consistency and reproducibility across different environments. Containers share the host operating system kernel but are isolated from each other through namespaces and control groups (cgroups), providing a level of abstraction and security. Docker containers can be started, stopped, moved, and deleted quickly and easily. They provide portability, scalability, and efficiency, allowing developers to package applications into standardized units that can run consistently on any infrastructure, from a developer's laptop to production servers or cloud environments. Containers are ephemeral,

meaning they can be spun up, used, and destroyed as needed, making them ideal for modern, dynamic, and scalable application architectures.

```
Item2 Shell Edit View Session Scripts Profiles Window Help
mdmoinuddinquazi@Moins-Macbook:~
Last login: Fri Apr 4 15:54:44 on ttys001
/Users/mdmoinuddinquazi/.zshenv:1:1 no such file or directory: /Users/mdmoinuddinquazi/.cargo/env

+ ~ docker --version
Docker version 28.0.1, build 068a01e

+ ~ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
5b7151e9710: Pull complete
Digest: sha256:722978494565d37d1262630108ab308d3e9ec7ed1c3286a32fe09856619a782
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's next:
View a summary of image vulnerabilities and recommendations • docker scout quickview ubuntu

+ ~ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
jakobsk/hetalex latest f7f4dc475ed7 5 weeks ago 23MB
ubuntu latest c3d1a3432580 2 months ago 101MB
collaborative-code-editor-client latest 434f595c25eb 2 months ago 1.31GB
collaborative-code-editor-worker latest 9ffdf69324d 2 months ago 1.15GB
collaborative-code-editor-server latest fcd689e2a29 2 months ago 1.15GB
redis latest a00436236602 2 months ago 140MB
mongo latest 0cb8696614c 3 months ago 849MB
<none> <none> feb4e6921a28 5 months ago 1.31GB
maink1309/collaborative-code-editor-worker v1 348e58cf060b 5 months ago 1.15GB
maink1309/collaborative-code-editor v1 1ceb984eb2aa 5 months ago 1.15GB
maink1309/collaborative-code-editor <none> 88407476b39f 5 months ago 1.15GB
mongodb/mongod-atlas-local 8.0 24064e551840 6 months ago 950MB
mongo <none> ba14dbfafa0b 6 months ago 847MB
gcr.io/k8s-minikube/kicbase v0.0.45 8411aacdb1cb 7 months ago 1.19GB
my-portfolio latest 576cfff65605 8 months ago 1.13GB
<none> <none> 18a6828bfff1 8 months ago 1.3GB
collaborative-code-editor-master-worker latest 9c8c37452ae4 8 months ago 1.15GB
collaborative-code-editor-master-server latest b43cd422c6a1 8 months ago 1.15GB
<none> <none> c74c95a2b075 8 months ago 1.15GB
<none> <none> 3d96354f9509 8 months ago 1.15GB
<none> <none> 697848ea38c 8 months ago 1.15GB
<none> <none> 8b1bb8b025ad 8 months ago 1.15GB
collaborative-code-editor-master-client latest 38fcc262e74b 8 months ago 1.26GB
<none> <none> fb1fa88be46 8 months ago 1.15GB
<none> <none> af5f47a5f902 8 months ago 1.15GB
<none> <none> 912f9e06a13f 8 months ago 1.15GB
<none> <none> ab5159c37362 8 months ago 1.15GB
image_tag latest 0f580830a015 9 months ago 1.15GB
backend_live latest 8a88bd498bd8 9 months ago 1.28GB
<none> <none> 21d8f4a26a8a 9 months ago 1.28GB
<none> <none> 5b4dae90baaa 9 months ago 1.28GB
<none> <none> 468d3f85dbc0 9 months ago 1.28GB
nginx latest 443d199eb0fc 9 months ago 193MB
mongo <none> 65b08ca2cb4f 10 months ago 762MB
```

```
Item2 Shell Edit View Session Scripts Profiles Window Help
root@894bcf74c86e: /
root@894bcf74c86e: / (docker)

maink1309/collaborative-code-editor v1 1ceb984eb2aa 5 months ago 1.15GB
maink1309/collaborative-code-editor <none> 88407476b39f 5 months ago 1.15GB
mongodb/mongod-atlas-local 8.0 24064e551840 6 months ago 950MB
mongo <none> ba14dbfafa0b 6 months ago 847MB
gcr.io/k8s-minikube/kicbase v0.0.45 8411aacdb1cb 7 months ago 1.19GB
my-portfolio latest 576cfff65605 8 months ago 1.13GB
<none> <none> 18a6828bfff1 8 months ago 1.3GB
collaborative-code-editor-master-worker latest 9c8c37452ae4 8 months ago 1.15GB
collaborative-code-editor-master-server latest b43cd422c6a1 8 months ago 1.15GB
<none> <none> c74c95a2b075 8 months ago 1.15GB
<none> <none> 3d96354f9509 8 months ago 1.15GB
<none> <none> 697848ea38c 8 months ago 1.15GB
<none> <none> 8b1bb8b025ad 8 months ago 1.15GB
collaborative-code-editor-master-client latest 38fcc262e74b 8 months ago 1.26GB
<none> <none> fb1fa88be46 8 months ago 1.15GB
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image_tag latest 0f580830a015 9 months ago 1.15GB
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<none> <none> 21d8f4a26a8a 9 months ago 1.28GB
<none> <none> 5b4dae90baaa 9 months ago 1.28GB
<none> <none> 468d3f85dbc0 9 months ago 1.28GB
nginx latest 443d199eb0fc 9 months ago 193MB
mongo <none> 65b08ca2cb4f 10 months ago 762MB
redis <none> e48e4ac2f06b 10 months ago 139MB
kindest/node <none> c67167dbf296 10 months ago 980MB
postgres latest f91377b20r11 11 months ago 453MB
gcr.io/k8s-minikube/kicbase v0.0.44 648c6eff48e4 11 months ago 1.17GB
docker/welcome-to-docker latest 648f93a1ba7d 17 months ago 19MB

+ ~ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES

+ ~ docker run it ubuntu /bin/bash
Unable to find image 'it:latest' locally
^[[Adocker: Error response from daemon: pull access denied for it, repository does not exist or may require 'docker login': denied: requested access to the resource is denied

Run 'docker run --help' for more information

+ ~ docker run it ubuntu
Unable to find image 'it:latest' locally
docker: Error response from daemon: pull access denied for it, repository does not exist or may require 'docker login': denied: requested access to the resource is denied

Run 'docker run --help' for more information

+ ~ docker run ubuntu

+ ~ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES

+ ~ docker run -it ubuntu
root@894bcf74c86e: /
```

```
iTerm2 Shell Edit View Session Scripts Profiles Window Help
mdmoimuddinquazi@Moins-Macbook:~
Last login: Sat Apr 5 16:13:44 on ttys001
/Users/mdmoimuddinquazi/.zshenv.:1: no such file or directory: /Users/mdmoimuddinquazi/.cargo/env
dock

+ ~ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
6c8351f555cc   ubuntu   "/bin/bash"             6 seconds ago Up 5 seconds   bold_mendelev

+ ~ docker container rm 6c8351f555cc
Error response from daemon: cannot remove container "/bold_mendelev": container is running: stop the container before removing or force remove

+ ~ docker container stop 6c8351f555cc
6c8351f555cc

+ ~ docker container rm 6c8351f555cc
6c8351f555cc

+ ~ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES

+ ~ docker images
REPOSITORY          TAG         IMAGE ID      CREATED       SIZE
jokobsk/netalertx   latest      f7f4dc475ed7  5 weeks ago  232MB
ubuntu              latest      c3d1a3432580  2 months ago 101MB
collaborative-code-editor-client latest      9ffdf6b9324d  2 months ago 1.15GB
collaborative-code-editor-worker latest      fc468f9e2a39  2 months ago 1.15GB
redis              latest      d0043b234602  2 months ago 140MB
mongo              latest      0cb86896614c  3 months ago 840MB
<none>             <none>      f4b4e4021a28  5 months ago 1.11GB
moink1309/collaborative-code-editor v1         1ceb984eb2aa  5 months ago 1.15GB
moink1309/collaborative-code-editor-worker v1         348e58cf46a0  5 months ago 1.15GB
moink1309/collaborative-code-editor <none>      88407476b39f  5 months ago 1.15GB
mongodb/mongod-atlas-local 8.0        24064e55184b  6 months ago 950MB
mongo              <none>      ba34bdf6fca0  6 months ago 847MB
gcr.io/k8s-minikube/kicbose v0.0.45    8411a0ac81cb  7 months ago 1.19GB
my-portfolio       latest      576cffff5605  8 months ago 1.13GB
<none>             <none>      18a6828b9ff1  8 months ago 1.3GB
<none>             <none>      c74a59a59b75  8 months ago 1.15GB
collaborative-code-editor-master-worker latest      9c8c37452a04  8 months ago 1.15GB
collaborative-code-editor-master-server latest      b43cd422c6a1  8 months ago 1.15GB
<none>             <none>      3d96354f95b9  8 months ago 1.15GB
<none>             <none>      8b1bb8d825ad  8 months ago 1.15GB
<none>             <none>      697848eac38c  8 months ago 1.15GB
collaborative-code-editor-master-client latest      38fcc262e74b  8 months ago 1.26GB
<none>             <none>      af9f47a5f902  8 months ago 1.15GB
<none>             <none>      fb1fa880be46  8 months ago 1.15GB
<none>             <none>      912f9b06a13f  8 months ago 1.15GB
<none>             <none>      a05159c37362  8 months ago 1.15GB
image_tag          latest      0f589b30a015  9 months ago 1.15GB
backend_live       latest      8a08bd4c99bd8 9 months ago 1.28GB
<none>             <none>      21a8f4a26a8a  9 months ago 1.28GB
<none>             <none>      5a0b3ae9b0aaa  9 months ago 1.28GB
```

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iTerm2 Shell Edit View Session Scripts Profiles Window Help
mdmoimuddinquazi@Moins-Macbook:~
redis          latest      d0043b234602  2 months ago 140MB
mongo          latest      0cb86896614c  3 months ago 840MB
<none>         <none>      f4b4e4021a28  5 months ago 1.11GB
moink1309/collaborative-code-editor v1         1ceb984eb2aa  5 months ago 1.15GB
moink1309/collaborative-code-editor-worker v1         348e58cf46a0  5 months ago 1.15GB
moink1309/collaborative-code-editor <none>      88407476b39f  5 months ago 1.15GB
mongodb/mongod-atlas-local 8.0        24064e55184b  6 months ago 950MB
mongo          <none>      ba34bdf6fca0  6 months ago 847MB
gcr.io/k8s-minikube/kicbose v0.0.45    8411a0ac81cb  7 months ago 1.19GB
my-portfolio   latest      576cffff5605  8 months ago 1.13GB
<none>         <none>      18a6828b9ff1  8 months ago 1.3GB
<none>         <none>      c74a59a59b75  8 months ago 1.15GB
collaborative-code-editor-master-worker latest      9c8c37452a04  8 months ago 1.15GB
collaborative-code-editor-master-server latest      b43cd422c6a1  8 months ago 1.15GB
<none>         <none>      3d96354f95b9  8 months ago 1.15GB
<none>         <none>      8b1bb8d825ad  8 months ago 1.15GB
<none>         <none>      697848eac38c  8 months ago 1.15GB
collaborative-code-editor-master-client latest      38fcc262e74b  8 months ago 1.26GB
<none>         <none>      af9f47a5f902  8 months ago 1.15GB
<none>         <none>      fb1fa880be46  8 months ago 1.15GB
<none>         <none>      912f9b06a13f  8 months ago 1.15GB
<none>         <none>      a05159c37362  8 months ago 1.15GB
image_tag      latest      0f589b30a015  9 months ago 1.15GB
backend_live   latest      8a08bd4c99bd8 9 months ago 1.28GB
<none>         <none>      21a8f4a26a8a  9 months ago 1.28GB
<none>         <none>      5a0b3ae9b0aaa  9 months ago 1.28GB
<none>         <none>      4e6d3f85e0cb  9 months ago 1.28GB
nginx          latest      443d199eb8fc  9 months ago 193MB
mongo          <none>      65bb8ca2cb4f  10 months ago 762MB
redis          <none>      e48e4ac2fa0b  10 months ago 139MB
kindest/node   latest      c67167dbf296  10 months ago 980MB
postgres       latest      f93377b220f1  11 months ago 453MB
gcr.io/k8s-minikube/kicbose v0.0.44    648c6affc8e4  11 months ago 1.17GB
docker/welcome-to-docker latest      648f93a1ba7d  17 months ago 19MB

+ ~ docker image rm c3d1a3432580
Error response from daemon: conflict: unable to delete c3d1a3432580 (must be forced) - image is being used by stopped container 255e1ab58309

+ ~ docker image rm c3d1a3432580
Error response from daemon: conflict: unable to delete c3d1a3432580 (must be forced) - image is being used by stopped container 894bcf74c86e

+ ~ docker rm 894bcf74c86e
894bcf74c86e

+ ~ docker image rm c3d1a3432580
Error response from daemon: conflict: unable to delete c3d1a3432580 (must be forced) - image is being used by stopped container 255e1ab58309

+ ~ docker image rm --force c3d1a3432580
Untagged: ubuntu:latest
Untagged: ubuntu/sha256:72297848456d5d37d1267630188ab3883e9ec7ed1c3286a32fe09856619a782
Deleted: sha256:c3d1a34325805c22bf44a5137224b0cf558dc6d885838c7746a6a2ea4eb191c

+ ~
```

B.3 Conclusion:

(Students must write the conclusion as per the attainment of individual outcome listed above and learning/observation)

Through this experiment, I gained a comprehensive understanding of containerization using Docker. I learned about the key differences between virtual machines and containers, the benefits of containerization, and how to implement Docker for application deployment. This experience enhanced my knowledge of software deployment and resource utilization in cloud computing environments.